

Question 11

Create a dataframe of ten rows, four columns with random values. Convert some values to nan values. Write a Pandas program which will highlight the nan values.

Aim

To create a DataFrame with 10 rows and 4 columns containing random values, convert some values to **NaN**, and highlight the **NaN** values using Pandas Styler.

Algorithm

1. Import the Pandas and NumPy libraries.
2. Create a DataFrame with 10 rows and 4 columns of random values.
3. Replace selected values with NaN.
4. Define a function to highlight NaN values.
5. Apply the style using DataFrame.style.applymap().
6. Display the styled DataFrame.

Input:

A,B,C,D
45,12,78,65
91,33,24,58
76,55,19,41
55,28,64,83
18,92,37,29
66,14,51,73
88,74,96,32
27,43,57,68
39,21,81,17
52,69,46,95

Code

```

D:\SANIA > cd D:\unit 2.py > ...
1  import pandas as pd
2  import numpy as np
3  # Read dataset
4  df = pd.read_csv(r"D:\SANIA\A,B,C,D.txt")
5  # Convert some values to NaN
6  df.iloc[2,1] = np.nan
7  df.iloc[5,3] = np.nan
8  df.iloc[7,0] = np.nan
9  # Highlight NaN values
10 def highlight_nan(value):
11     if pd.isna(value):
12         return "background-color:red;color:white"
13     return ""
14 styled_df = df.style.applymap(highlight_nan)
15 print(df)
16 styled_df

```

Output:

```

        styled_df = df.style.applymap(highlight_nan)

```

	A	B	C	D
0	45.0	12.0	78	65.0
1	91.0	33.0	24	58.0
2	76.0	NaN	19	41.0
3	55.0	28.0	64	83.0
4	18.0	92.0	37	29.0
5	66.0	14.0	51	NaN
6	88.0	74.0	96	32.0
7	NaN	43.0	57	68.0
8	39.0	21.0	81	17.0
9	52.0	69.0	46	95.0

Result

The DataFrame is created successfully. Selected values are converted to **NaN**, and the **NaN cells are highlighted in red with white text.**

Question 12

Aim

To create a DataFrame with 10 rows and 4 columns containing random values and apply **black background color** with **yellow font color** using Pandas Styler.

Algorithm

1. Import the Pandas and NumPy libraries.
2. Create a DataFrame with random values.
3. Use the `style.set_properties()` method.
4. Set the background color to black.
5. Set the text color to yellow.
6. Display the styled DataFrame.

```
1  import pandas as pd
2  # Read dataset
3  df = pd.read_csv(r"D:\SANIA\A,B,C,D.txt")
4  # Apply black background and yellow text
5  styled_df = df.style.set_properties(**{
6      "background-color": "black",
7      "color": "yellow"
8  })
9  print(df)
10 styled_df
```

Output:

	A	B	C	D
0	21	75	33	64
1	58	19	87	42
2	96	51	23	11
3	67	84	45	39
4	14	28	76	93
5	52	61	35	80
6	90	47	18	72
7	30	66	99	54
8	74	25	40	88
9	17	82	63	29